A BOUNDED FILTERED COMPLEX IS RESOLVED BY FILTERED COMPLEXES IN THE PRESCRIBED RANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Filtered complex and its associated graded complexes. defined in the same section, source lines 16330-16420, and used throughout the proof. Supplies the object of the statement and the shape of its filtration.
- A module admits a resolution by free or projective modules, and such a resolution lifts along a filtration. used at proof lines 15670-15720 to produce the next stage of the induction. Provides the complex that is glued on at each step.
- The long exact sequence of cohomology attached to a short exact sequence of complexes. used to compare the cohomology of consecutive stages at proof lines 15700-15760. Shows that each stage is quasi-isomorphic to the original complex.

Source reference. Original lemma 0H7X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21472–21472. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06Y3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8872–8872. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BYH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15015–15015. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a filtered complex. There exists a map $f : P^\bullet \to K^\bullet$ of filtered complexes such that*

1. *each P^n, F^iP^n, gr^iP^n is a free R-module,*
2. *the complexes of R-modules $P^\bullet$, $F^iP^\bullet$, and $gr^iP^\bullet$ are K-flat,*
3. *f induces quasi-isomorphisms $P^\bullet \to K^\bullet$, $F^iP^\bullet \to F^iK^\bullet$, and $gr^iP^\bullet \to gr^iK^\bullet$.*

Proof. Let us say a filtered complex $L^\bullet$ is *basic* if each L^n, F^iL^n, gr^iL^n is a free R-module and if all differentials are zero.

There exists a basic filtered complex $P_0^\bullet$ and a map $f_0 : P_0^\bullet \to K^\bullet$ of filtered complexes such that f_0 and F^if_0, $i \in \mathbf{Z}$ are surjective on cohomology. To see this set

$$P_0^n = \bigoplus\nolimits_{z \in \mathrm{Ker}(d_K^n)} R\xi_z \oplus \bigoplus\nolimits_{j \in \mathbf{Z}} \bigoplus\nolimits_{z \in \mathrm{Ker}(d^n_{F^jK})} R\xi_z$$

with zero differentials and the map f_0 defined by $f_0(\xi_z) = z$. As for the filtration, we set

$$F^iP_0^n = \bigoplus\nolimits_{j \geq i \in \mathbf{Z}} \bigoplus\nolimits_{z \in \mathrm{Ker}(d^n_{F^jK})} R\xi_z$$

We leave it to the reader to check that this gives $f_0 : P_0^\bullet \to K^\bullet$ as claimed.

By induction on $m \geq 0$ we are going to construct embeddings

$$P_0^\bullet \subset \ldots P_m^\bullet \subset P_{m+1}^\bullet$$

of filtered complexes and maps $f_m : P_m^\bullet \to K^\bullet$ with the following properties

1. the filtered complex $P_{m+1}^\bullet/P_m^\bullet$ is basic,
2. the kernel of $H^n(f_m)$ and $H^n(F^if_m)$ maps to zero in $H^n(P_{m+1}^\bullet)$ and $H^n(F^iP_{m+1}^\bullet)$,
3. the map $f_{m+1} : P_{m+1}^\bullet \to K^\bullet$ extends the map f_m.

To to this, set

$$\Omega_{n,m} = \mathrm{Ker}\left(\mathrm{Ker}(d^n_{P_m}) \to H^n(K^\bullet)\right), \text{ resp. } \Omega_{n,i,m} = \mathrm{Ker}\left(\mathrm{Ker}(d^n_{F^iP_m}) \to H^n(F^iK^\bullet)\right),$$

Note that $\Omega_{n,m}$ surjects onto the kernel of $H^n(f_m)$ and that $\Omega_{n,i,m}$ surjects onto the kernel of $H^n(F^if_n)$. For each $z \in \Omega_{n,m}$, resp. $z \in \Omega_{n,i,m}$ we choose a $y_z \in K^{n-1}$, resp. $y_z \in F^iK^{n-1}$ such that $d_K(y_z) = f_m(z)$. Then we set

$$P_{m+1}^n = P_m^n \oplus \bigoplus\nolimits_{z \in \Omega_{n+1,m}} R\eta_z \oplus \bigoplus\nolimits_{j \in \mathbf{Z}} \bigoplus\nolimits_{z \in \Omega_{n+1,j,m}} R\eta_z$$

We set $d_{P_m}(\eta_z) = z$ and we set $f_{m+1}(\eta_z) = y_z$. Finally, we set

$$F^iP_{m+1}^n = F^iP_m^n \oplus \bigoplus\nolimits_{j \geq i} \bigoplus\nolimits_{z \in \Omega_{n+1,j,m}} R\eta_z$$

We leave it to the reader to check that this gives $P_m^\bullet \subset P_{m+1}^\bullet$ and $f_{m+1} : P_{m+1}^\bullet \to K^\bullet$ as claimed.

At this point we simply take

$$P^\bullet = \bigcup P_m^\bullet$$

as a filtered complex with map $f : P^\bullet \to K^\bullet$ given by $\bigcup f_m$.

Part (1) of the statement of the lemma holds because P^n as a filtered module is isomorphic to the direct sum of P_0^n and P_{m+1}^n/P_m^n for $m \geq 0$. Small detail omitted.

Part (3) of the statement. Observe that $H^n(P^\bullet) = \mathrm{colim}\, H^n(P_m^\bullet)$. The map f_0 is surjective on cohomology; whence the same holds for each f_m. For each m by construction the embedding $P_m^\bullet \subset P_{m+1}^\bullet$ kills the kernel of $H^n(f_m)$. Combining

these facts the reader easily concludes that $H^n(f)$ is an isomorphism. Similarly for $H^n(F^i f_n)$. Then also $H^n(\mathrm{gr}^i f)$ must be an isomorphism because of the short exact sequence $0 \to F^{i+1} \to F^i \to \mathrm{gr}^i \to 0$ (of functors on the category of filtered complexes, say). Small detail omitted.

Part (2) of the statement. To see that $P^\bullet$ is K-flat, by Lemma 06Y3, it suffices to show that $P^\bullet_m$ is K-flat. By Lemma 0BYH and induction it suffices to note that a complex with zero differentials and free terms is K-flat. The same argument works to show that $F^i P^\bullet$ is K-flat for all $i \in \mathbf{Z}$. Finally, we see that $\mathrm{gr}^i P^\bullet$ is K-flat by another application of Lemma 0BYH. $\square$